Module7_lab2

October 18, 2026

[1]:
```
import pandas as pd
```

[2]:
```
from sklearn.linear_model import LinearRegression
```

[3]:
```
df = pd.read_csv("Downloads/ratings.tsv",sep="\t")
```

[4]:
```
df.shape
```

[4]:
```
(15, 5)
```

[5]:
```
df.head()
```

[5]:

	restaurant	food	ambience	service	rating
0	1	85	82	89	78
1	2	80	90	80	85
2	3	83	86	83	85
3	4	70	96	75	72
4	5	68	80	78	75

[6]:
```
df.describe()
```

[6]:

	restaurant	food	ambience	service	rating
count	15.000000	15.000000	15.000000	15.000000	15.000000
mean	8.000000	73.133333	81.733333	75.133333	73.066667
std	4.472136	6.555986	9.786774	8.069932	8.319226
min	1.000000	64.000000	68.000000	56.000000	54.000000
25%	4.500000	68.500000	71.000000	73.500000	70.500000
50%	8.000000	72.000000	82.000000	78.000000	74.000000
75%	11.500000	77.500000	90.000000	78.000000	77.500000
max	15.000000	85.000000	96.000000	89.000000	85.000000

[7]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   restaurant  15 non-null     int64
```

```
 1   food        15 non-null     int64
 2   ambience    15 non-null     int64
 3   service     15 non-null     int64
 4   rating      15 non-null     int64
dtypes: int64(5)
memory usage: 732.0 bytes
```

[8]:
```
x=df[["food","service","ambience"]]
y=df["rating"]
```

[9]:
```
model=LinearRegression()
model.fit(x,y)
```

[9]:
```
LinearRegression()
```

[10]:
```
print(f"Intercept: {model.intercept_:.4f}\n")
for feature, coef in zip(x.columns, model.coef_):
  print(f"  {feature}: {coef:.4f}")
```

```
Intercept: -15.8713

  food: 0.7338
  service: 0.3240
  ambience: 0.1338
```

A struggling restaurant should fix its food quality first, as it has by far the largest positive coefficient at 0.7338 and drives customer ratings more than the other 2 factors of sevice and ambience. While service and ambience matter, investing into the food will give the highest return on overall customer satisfaction.

[ ]:
```

```